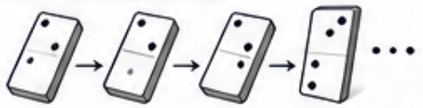
COMP233 — Discrete Mathematics

# Lecture 5.2–5.3 Mathematical Induction I & II

*A Proof Technique for Infinite Families of Statements*

## 1. Motivation & Intuition

- How can we prove a statement $P(n)$ for infinitely many integers $n$?
- **Idea:** a domino chain.



$P(1) \to P(2) \to P(3) \to \cdots$

## 2. Principle of Mathematical Induction

**Theorem** (Principle of MI)

Let $P(n)$ be a statement and let $a \in \mathbb{Z}$. If

(1) $P(a)$ is true, and

(2) for every $k \geq a$, $P(k) \to P(k+1)$ is true,

then $P(n)$ is true for all $n \geq a$.

## 3. Key Terms

| | |
|---|---|
| $P(n)$ | A statement about an integer $n$. |
| **Basis step** | Prove $P(a)$ (the starting case). |
| **Inductive hypothesis (IH)** | Assume $P(k)$ is true for an arbitrary $k \geq a$. |
| **Inductive step** | Using the IH, prove $P(k+1)$. |
| **Conclusion** | $P(n)$ is true for all $n \geq a$. |

## 4. Induction vs. Inductive Reasoning

| | Inductive Reasoning | Mathematical Induction |
|---|---|---|
| **Purpose** | Suggests a conjecture (pattern). | Proves the statement for all $n \geq a$. |
| **Form** | $P(1), P(2), \ldots \Rightarrow$ conjecture $P(n)$. | $P(a)$ and $P(k) \to P(k+1) \Rightarrow P(n)$. |
| **Status** | Not a proof (may fail). | A valid proof (theorem). |

## 5. The Induction Proof Template

**Claim:** $\forall n \geq a$, $P(n)$ is true.

1. **Basis Step** — Prove $P(a)$.
2. **Inductive Step** — Let $k \geq a$ be arbitrary. Assume $P(k)$ is true (IH). Show that $P(k+1)$ is true.
3. **Conclusion** — Therefore, by mathematical induction, $P(n)$ is true for all $n \geq a$.

## 6. How to Construct an Induction Proof

**Start**

↓ Identify $P(n)$ and the starting index $a$.

↓ Prove the basis step $P(a)$.

↓ Assume $P(k)$ (IH).

↓ Work from the LHS of $P(k+1)$, use the IH to rewrite, and prove $P(k+1)$.

↓ **Conclude:** $\forall n \geq a$, $P(n)$.

## 7. Common Mistakes and Fixes

| Typical Error (Invalid) | Correction / Key Message |
|---|---|
| Trying to prove $P(k)$. | $P(k)$ is assumed (IH), not proved. |
| Checking several cases (e.g., $n = 1, 2, 3, \ldots$) and claiming a proof. | It only suggests a pattern; it is not a proof for all $n$. |
| Writing an incorrect $P(k+1)$ (e.g., only replacing the last $k$). | Write $P(n)$ first, then substitute $n = k+1$ completely. |
| Not verifying the basis step ("obviously true"). | Show explicitly that $P(a)$ is true. |
| Not using the IH in the step. | The inductive step must use $P(k)$. |
| Assuming an inappropriate starting index. | The starting index must match the statement (e.g., $a = 0$ or 1). |

## 8. Worked Examples by Mathematical Induction

### (A) Sum of the First $n$ Integers

Prove $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$ for all $n \geq 1$.

- **Basis ($n = 1$):** $1 = \dfrac{1(2)}{2}$ ✓
- **IH:** Assume $P(k)$: $1 + \cdots + k = \dfrac{k(k+1)}{2}$.
- **Step:**

$$1 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+\ ) = \frac{(k+\ )(k+2)}{2}.$$

∴ Therefore, $P(n)$ is true.

### (B) Sum of Odd Integers

Prove $1 + 3 + \cdots + (2n-1) = n^2$ for all $n \geq 1$.

- **Basis ($n = 1$):** $1 = 1^2$ ✓
- **IH:** $P(k)$: $1+3+\cdots+(2k-1) = k^2$.
- **Step:**

$$1+3+\cdots+(2k-1)+(2k+1) = k^2 + (2k+1) = (k+1)^2.$$

∴ Therefore, $P(n)$ is true.

### (C) Geometric Sum

Prove $1 + r + r^2 + \cdots + r^n = \dfrac{r^{n+1}-1}{r-1}$ for $r \neq 1$ and $n \geq 0$.

- **Basis ($n = 0$):** $1 = \dfrac{r-1}{r-1}$ ✓
- **IH:** $P(k)$: $1 + \cdots + r^k = \dfrac{r^{k+1}-1}{r-1}$.
- **Step:**

$$1 + \cdots + r^k + r^{k+1} = \frac{r^{k+1}-1}{r-1} + r^{k+1} = \frac{r^{k+2}-1}{r-1}.$$

∴ Therefore, $P(n)$ is true.

### (D) Divisibility

Prove $3 \mid (2^{2n}-1)$ for all $n \geq 1$.

- **Basis ($n = 1$):** $2^2 - 1 = 3$ ✓
- **IH:** $2^{2k} - 1 = 3r$ for some $r \in \mathbb{Z}$.
- **Step:**

$$\begin{aligned} 2^{2(k+1)} - 1 &= 4 \cdot 2^{2k} - 1 \\ &= 3 \cdot 2^{2k} + (2^{2k} - 1) \\ &= 3 \cdot 2^{2k} + 3r \\ &= 3\,(2^{2k} + r). \end{aligned}$$

∴ Therefore, $3 \mid (2^{2n}-1)$.

### (E) Inequality

Prove $2^n \geq n+1$ for all $n \geq 0$.

- **Basis ($n = 0$):** $1 \geq 1$ ✓
- **IH:** $2^k \geq k+1$.
- **Step:**

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &\geq 2\,(k+1) \\ &\geq k+2. \end{aligned}$$

∴ Therefore, $P(n)$ is true.

## 9. Properties of Sequences (via Induction)

Induction can prove properties of sequences defined recursively or explicitly.

**Example: Fibonacci numbers**

Let $F_0 = 0$, $F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$.

Claim: $F_k < 2^k$ for all $k \geq 0$.

**Proof:** (by induction with the recursive definition)

- **Basis:** $F_0 = 0 < 1 = 2^0$. • **IH:** $F_k < 2^0$.
- **Step:** $F_{k+1} = F_k + F_{k-1} < 2^k + 2^{k-1} < 2^{k+1}$.

## 10. Applications

- Proving correctness of iterative or **recursive** algorithms.
- Establishing **formulas** for sums and geometric sequences.
- Proving divisibility and **inequalities** for all integers.
- Analyzing properties of sequences (e.g., Fibonacci, bounds, growth).

## 11. Key Takeaways

- ☐ To prove $P(n)$ for all $n \geq a$:
  - Prove the basis $P(a)$.
  - Assume $P(k)$ and prove $P(k+1)$.
- ☐ The IH is an **assumption**, not a theorem.
- ☐ Start from the LHS of $P(k+1)$ and use the IH.
- ☐ Induction proves infinitely many cases.
- ☐ Checking examples alone is not a proof.

## 12. Example of an Invalid Proof (Counterexample)

**False claim:** $1 + 2 + \cdots + n = n^2$ (for all $n \geq 1$). ✗

**"Proof" (invalid):**

- **Basis:** $n = 1$: $1 = 1^2$.
- **IH:** Assume $1 + \cdots + k = k^2$.
- **Step (incorrect):** $1 + \cdots + (k+1) = k^2 + (k+1) = (k+1)^2$.

**Error:** The algebra is incorrect: $k^2 + (k+1) \neq (k+1)^2$.